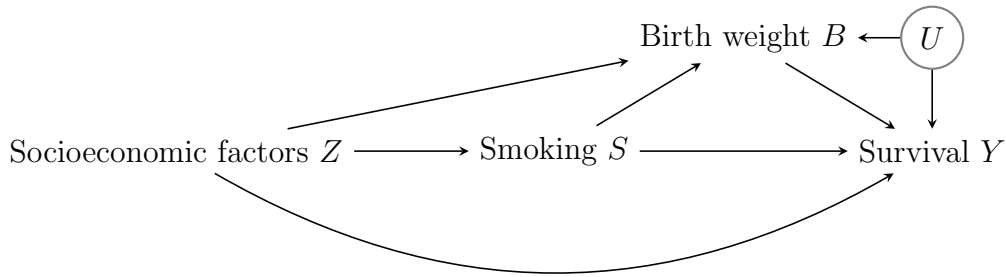


Practical 6: solutions

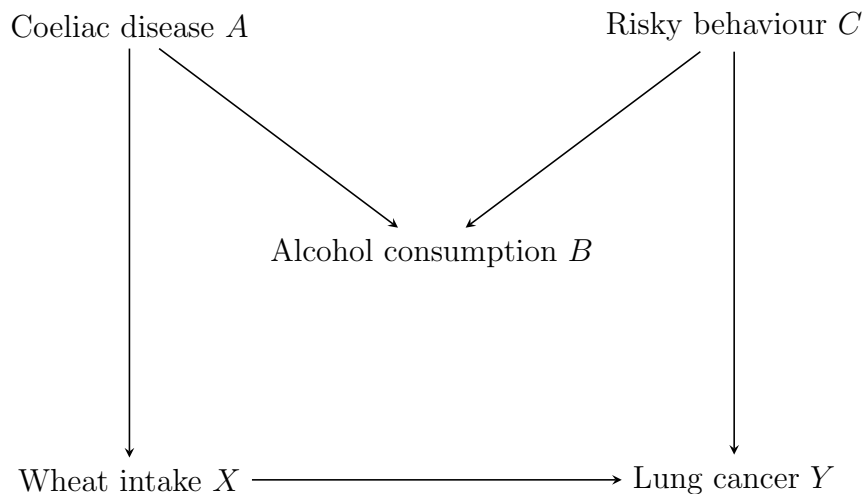
Part A: DAGs

1. Consider the smoking and birth weight example from the lecture in Section 6.3.4, and the DAG in Figure 6.2. Suppose you wanted to estimate the causal effect of birth weight on survival. Which, if any, of the other variables in the DAG would you need to adjust for to do this?



To estimate the causal effect of birth weight on survival we would need to control for smoking S , socioeconomic factors Z , and the unmeasured variable U . Because U is unmeasured we would therefore not be able to control for all confounders. We would then need to consider the possible impact of not adjusting for U , and could try to identify some measured variables that could serve as partial proxies for U .

2. In another study it is of interest to estimate the causal effect of wheat intake on risk of lung cancer. The DAG is shown below, and all variables depicted in the DAG are measured in the observed data. Would you need to adjust for alcohol consumption to estimate the causal effect of wheat intake on lung cancer?



In this DAG there are two paths from X to Y . One is the causal path $X \rightarrow Y$, and the other is the non-causal path $X \leftarrow A \rightarrow B \leftarrow C \rightarrow Y$. The path $X \leftarrow A \rightarrow B \leftarrow C \rightarrow Y$ is blocked by the collider B , so it is a closed path.

We do not need to control for any of the variables A, B, C in order to control for confounding. If we controlled for B then we would in fact open up the path $X \leftarrow A \rightarrow B \leftarrow C \rightarrow Y$. Then to block that path we would additionally need to control for A or C (or both).

Part B: Variable selection strategies

In this part we will use a data set containing information on the birth weight of 4642 babies, alongside information about the baby's mother and her maternal history, and some information about the baby's father. The aim is to estimate the causal effect of maternal smoking (yes or no) on the risk of low birth weight, defined as birth weight ≤ 2500 g. The data were described in a paper by Cattaneo 2010 (Journal of Econometrics 155: 138-154), and are available directly from Stata: `use https://www.stata-press.com/data/r16/cattaneo2, clear`

1. Explore the dataset.
 - (a) Identify which variables correspond to the outcome Y , exposure X and set of potential confounders Z . Are there are variables you think should not be included in Z ?
 - (b) One issue to look for is where there are very few observations in a particular level of a categorical variable. If there are, we can merge categories to avoid running into problems during regression analysis. Are there any variables you would consider re-categorising in this data?
- (a) The outcome variable Y is `lbweight` and the exposure variable X is `mbsmoke`. The following variables should be excluded from Z because they are alternative measures of the exposure and outcome: `bweight` and `msmoke`. The variable `fbaby` (indicator of first baby) is a cruder version of the variable `order` (birth order), so we can exclude `fbaby` as the same information is contained in the `order` variable. The variables `prenatal` (trimester of first prenatal care visit) and `prenatal1` (indicator of whether the first prenatal care visit was in first trimester) could be on the causal pathway from the exposure to the outcome, and so we decided to exclude these. The variables included in Z are therefore: `mmarried, mhispanic, fhispanic, foreign, alcohol, deadkids, mage, medu, fage, fedu, monthslb, order, mrace, frace, birthmonth`.
- (b) The `order` variable has quite a few categories that have a very small number of individuals. We recoded this into the categories: 1, 2, ≥ 3 . Those recorded as 0 were included in category 1.
2. Summarise the exposure and outcome. Investigate the association between the mother's smoking status and risk of low birth weight, without adjustment for any other variables.

When looking at the crude association, the proportion of mothers whose baby had low birth weight was lower among non-smokers (4.9%) than among smok-

ers (11%). Performing a logistic regression of `lbweight` on `mbsmoke` gives an odds ratio estimate for `mbsmoke` of 2.40 (95% CI 1.85 to 3.11). The odds of a low birth weight baby in smokers is 2.4 times the odds of a low birth weight baby in non-smokers. The 95% confidence interval excludes 1 (by some way), so there is strong evidence of an association between the mother's smoking status and low birth weight (p-value < 0.001).

3. Fit an appropriate regression model for the outcome Y given X and the full set of potential confounders Z , and interpret the coefficient for X . What is the estimated marginal mean difference $E(Y|do(X = 1)) - E(Y|do(X = 0))$, and what assumptions do you make to obtain this? Interpret your results.

A logistic regression model was fitted with outcome `lbweight` and explanatory variables `mbsmoke` (the main exposure) and Z (the set of variables listed in 1(a)). The estimated odds ratio for `mbsmoke` is 2.09 (95% CI 1.57 to 2.78). This is conditional on the other variables in the model. Conditional on Z the odds of a low birth weight baby in smokers is 2.09 times the odds of a low birth weight baby in non-smokers. The 95% confidence interval excludes 1 (by some way), so there is strong evidence of an association between the mother's smoking status and low birth weight (p-value < 0.001) conditional on Z .

The estimated marginal risk difference is 0.047 (95% CI 0.024 to 0.069). The interpretation is that if all of the mothers had been smokers the risk of a low birth weight baby would have been 4.7 percentage points higher than if all of the mothers had been non-smokers. We have made the assumption that the set of adjust variables Z included in the logistic model is sufficient to control for confounding, i.e. that there is no unmeasured confounding.

Note that it might be more sensible to consider how much the risk of low birth weight would have been reduced among smokers if they had given up smoking - this would be an example of estimating the average effect of exposure in the exposed, rather than overall average effect of exposure.

4. Perform forwards selection using a p-value for inclusion of $p^* = 0.1$. What set of adjustment variables do you end up with? Using this set of variables, use the standardisation method to obtain an estimate of the marginal mean difference, $E(Y|do(X = 1)) - E(Y|do(X = 0))$, and corresponding confidence interval.

For categorical variables we don't want each of the binary 'dummy' variables to be treated as separate variables in the selection process. They should be considered collectively. To force Stata to include or exclude all categories together you should enclose the variable in brackets in the command statement.

In this method the main exposure variable is forced into the model. The forwards selection approach selects only two of the Z variables: `mrace` and `mmarried`. After the selection procedure we fitted the logistic regression model

for `lbweight` including `mbsmoke`, `mrace` and `mmarried`, and then used standardisation. The estimated marginal risk difference is 0.048 (95% CI 0.027 to 0.068). The estimate is very similar to that obtained from the full model in question 3. However, the confidence interval is a little narrower.

5. Perform the change-in-estimates method using the forwards selection approach. What set of adjustment variables do you end up with? Using this set of variables what is the estimated marginal mean difference, $E(Y|do(X = 1)) - E(Y|do(X = 0))$, and corresponding confidence interval.

We used the change in estimates method in which we add variables if they give a change of 10% or more in the coefficient for `mbsmoke`. This resulted in first adding the variable `mmarried` and then adding the variable `mrace`. So we end up with the same set of adjustment variables as when we used the forwards selection approach in question 4. The estimated conditional OR and marginal mean difference are therefore the same.

6. Perform the double-selection approach. What set of adjustment variables do you end up with? Using this set of variables what is the estimated marginal mean difference, $E(Y|do(X = 1)) - E(Y|do(X = 0))$, and corresponding confidence interval. When using the `margins` command you can use the `vce(unconditional)` option to obtain standard errors and confidence intervals based on sandwich variance estimates.

- Step 1. The first step of double-selection involves performing forwards selection based on a logistic regression for the outcome `lbweight` with candidate explanatory variables Z . The p-value for inclusion is $0.1/K \log(n) = 0.1/(15 * \log(4642)) = 0.00078962$, where K is the number of variables in Z and n is the sample size. This step selects `mmarried` and `mrace`.
- Step 2. The second step of double-selection involves performing forwards selection based on a logistic regression using the exposure of interest `mbsmoke` as the outcome variable, and with candidate explanatory variables Z . This step selects: `mmarried`, `alcohol`, `medu`, `monthslb`, `mrace`, `mhisp`, `prevnewbornddeath`, `fedu`.
- Step 3. The combined set of adjustment variables based on those selected either in step 1 or in step 2 happens to be the same as the set of variables selected in step 2 here (which included `mmarried` and `mrace` selected in step 1). This set of variables is denoted Z^* . The final step is to fit a logistic regression of `lbweight` on `mbsmoke` and Z^* (`mmarried`, `alcohol`, `medu`, `monthslb`, `mrace`, `mhisp`, `prevnewbornddeath`, `fedu`), which we fit using sandwich variance estimators. The estimated OR for `mbsmoke` conditional on Z^* is 2.05 (95% CI 1.54 to 2.75). Unlike in the single selection approach in questions 4 and 5, the confidence interval is valid.

After performing standardisation using the model from step 3, the estimated

marginal risk difference is 0.046 (95% CI 0.024 to 0.069). Again, unlike in the single selection approaches in questions 4 and 5, the confidence interval is valid.

7. Make a table presenting your results from the different methods and write an accompanying paragraph summarising the results. Include some explanation as to why you see different results from the different methods (if this is the case).

Table 1: Effect of the mother smoking on low birth weight. Results from the unadjusted model and four adjusted models based on different variable selection approaches.

Analysis strategy	Odds ratio (95% CI)*	Marginal risk difference (95% CI)**
Unadjusted	2.40 (1.85, 3.11)	0.061 (0.039, 0.083)
Full model	2.09 (1.57, 2.78)	0.047 (0.026, 0.069)
Forwards selection	2.09 (1.59, 2.74)	0.048 (0.027, 0.069)
Change-in-estimates	2.09 (1.59, 2.74)	0.048 (0.027, 0.069)
Double-selection	2.05 (1.54, 2.75)	0.046 (0.024, 0.069)

* Odds ratio from the unadjusted ("Unadjusted") or adjusted (all other methods) logistic regression of low birth weight on smoking status. ** Marginal risk difference for low birth weight in smokers vs non-smokers based on standardisation.

Example summary of the results (which assumes the methods have been described earlier):

The results are summarised in Table 1. Before adjustment for potential confounders, the odds of low birth weight in smokers was estimated to be 2.4 times that in non-smokers, and this association was highly statistically significant (OR 2.40, 95% CI 1.85 to 3.11). This corresponds to a risk difference of 0.061 (95% CI 0.039 to 0.083), meaning that the crude risk of a low birth weight baby was 6.1 percentage points higher in smokers compared to non-smokers. We identified 15 potential confounders. Fitting a logistic model for the outcome (low birth weight) given the exposure (mother smoking) with adjustment for all 15 potential confounders gave a conditional OR estimate of 2.09 (95% CI 1.57 to 2.78), and corresponding marginal risk difference (based on standardisation) of 0.047 (95% CI 0.026 to 0.069). The interpretation of the marginal risk difference is that if all of the mothers had been smokers the risk of a low birth weight baby would have been 4.7 percentage points higher than if all of the mothers had been non-smokers. This causal interpretation relies on the assumption that the set of adjustment variables included in the logistic regression model is sufficient to control for confounding. The forwards selection approach only selected two out of the 15 potential adjustment variables: mother's marital status and mother's ethnicity. The change-in-estimates procedure selected exactly the same two adjustment variables as the forwards selection approach and therefore gave identical estimates. The estimates obtained from the forwards selection and change-in-estimates methods were nearly identical

to those obtained from the full model, though the confidence intervals are slightly narrower. However, the standard confidence intervals following these single-selection methods are known not to be valid. The double-selection approach identified a larger set of 8 adjustment variables, which included the two identified using forwards selection and change-in-estimates. The double-selection approach gave a conditional OR estimate of 2.05 (95% CI 1.54 to 2.75), and corresponding marginal risk difference of 0.046 (95% CI 0.024 to 0.069). The confidence intervals obtained from the double-selection approach are valid.

8. EXTRA: In Question 4, try backwards selection instead of forwards selection. Do you get a different set of variables? Try varying the p-value for inclusion/exclusion in the forwards/backwards selection methods to see what impact this has.

See the code file for some examples of how the results change if we use a different p-value for inclusion, or if we use backwards elimination.